PORTFOLIO MANAGEMENT SYSTEM

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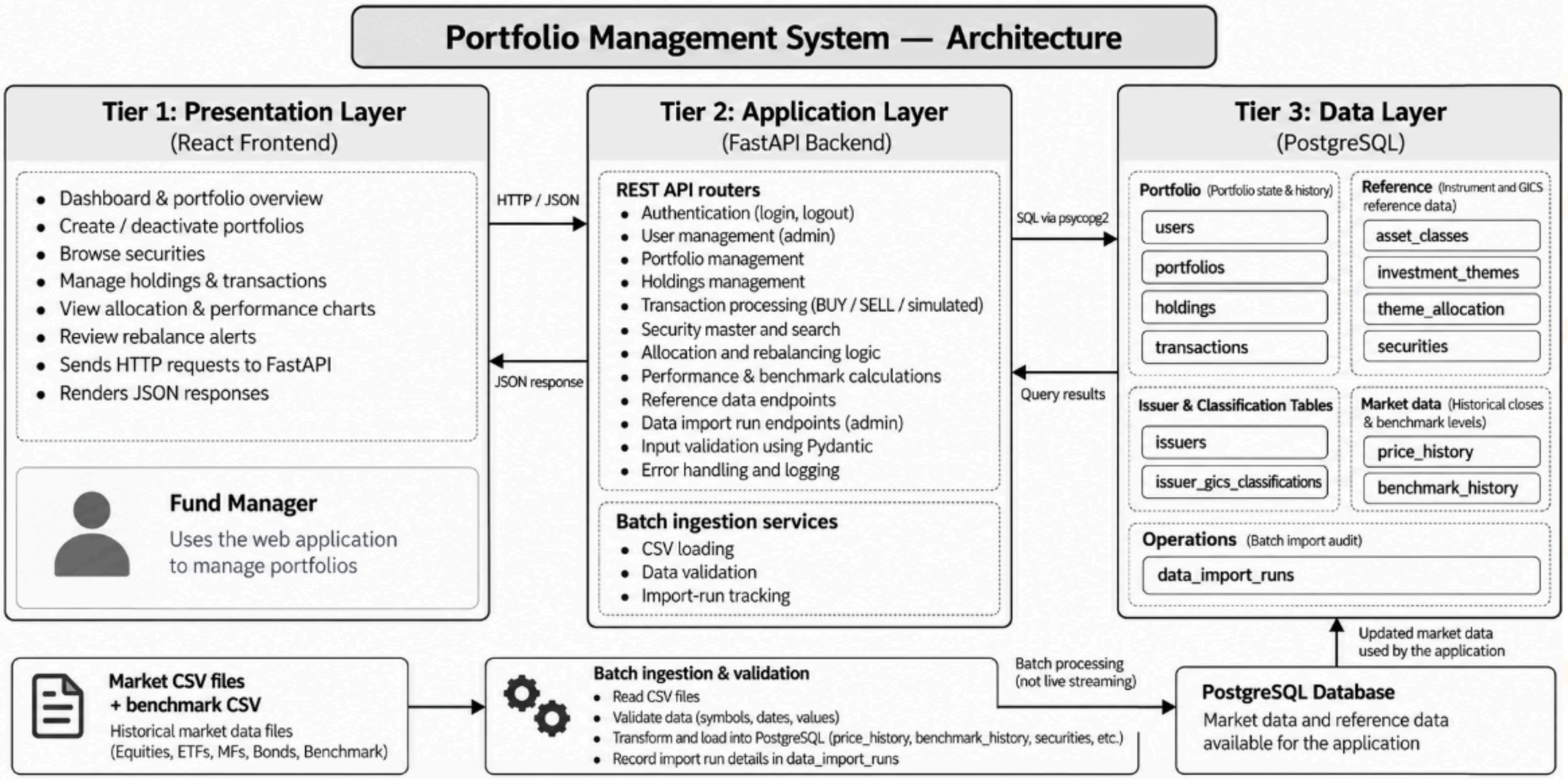
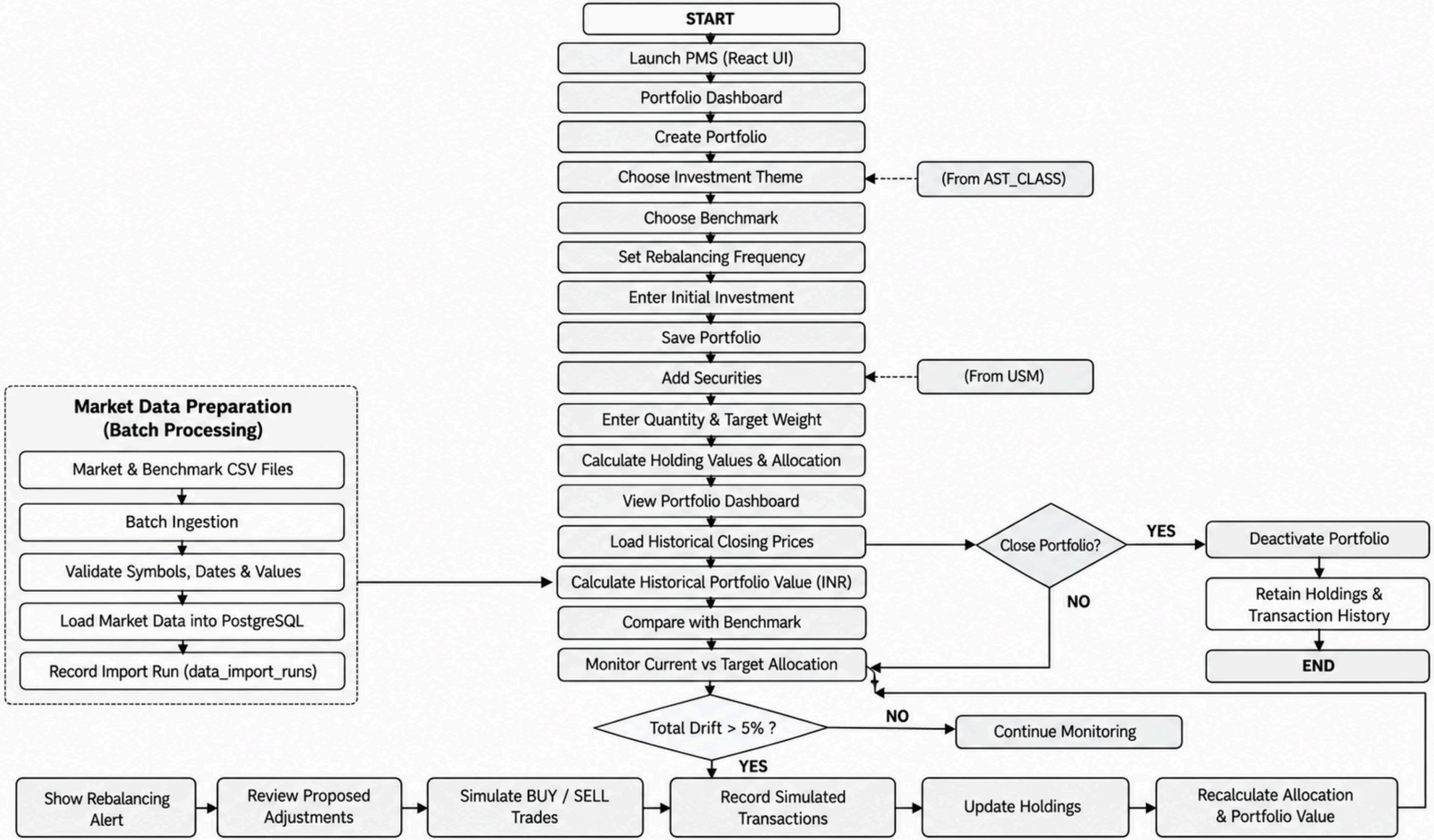
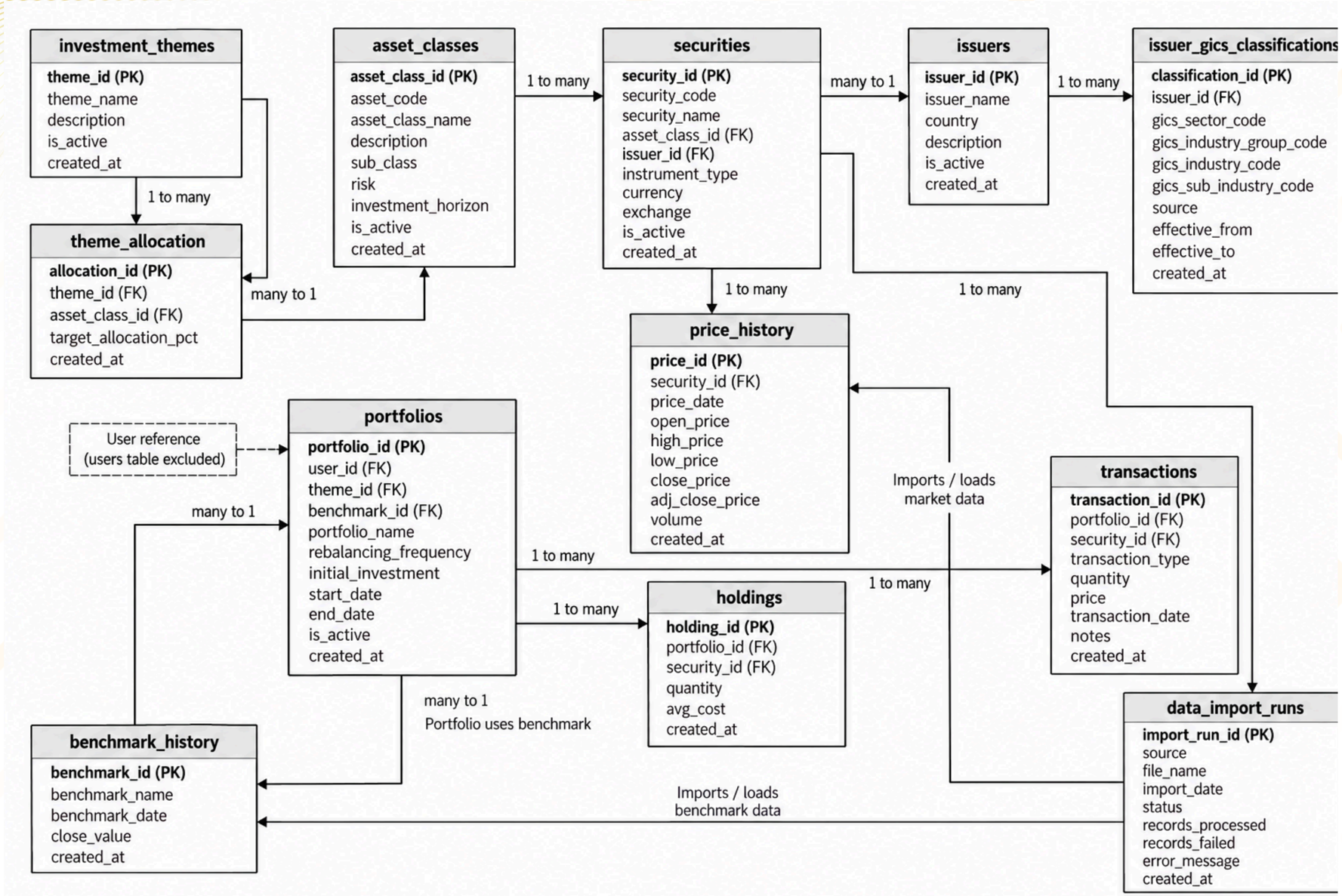
The Portfolio Management System is a financial application built using React, FastAPI, PostgreSQL. It integrates the Unified Security Master (USM) and Asset Class service to manage securities, asset classifications, portfolios, holdings, analytics, and historical market data. The system provides benchmark comparison, portfolio performance tracking, asset-class and rebalancing, simulated BUY/SELL execution, and portfolio alerts.

Portfolio Management System — Architecture



Portfolio Management System — Workflow




investment_themes
theme_id (PK)
theme_name
description
is_active
created_at
1 to many
theme_allocation
allocation_id (PK)
theme_id (FK)
asset_class_id (FK)
target_allocation_pct
created_at
many to 1
asset_classes
asset_class_id (PK)
asset_code
asset_class_name
description
sub_class
risk
investment_horizon
is_active
created_at
1 to many
securities
security_id (PK)
security_code
security_name
asset_class_id (FK)
issuer_id (FK)
instrument_type
currency
exchange
is_active
created_at
many to 1
issuers
issuer_id (PK)
issuer_name
country
description
is_active
created_at
1 to many
issuer_gics_classifications
classification_id (PK)
issuer_id (FK)
gics_sector_code
gics_industry_group_code
gics_industry_code
gics_sub_industry_code
source
effective_from
effective_to
created_at
1 to many
1 to many
price_history
price_id (PK)
security_id (FK)
price_date
open_price
high_price
low_price
close_price
adj_close_price
volume
created_at
User reference
(users table excluded)
portfolios
portfolio_id (PK)
user_id (FK)
theme_id (FK)
benchmark_id (FK)
portfolio_name
rebalancing_frequency
initial_investment
start_date
end_date
is_active
created_at
many to 1
1 to many
1 to many
holdings
holding_id (PK)
portfolio_id (FK)
security_id (FK)
quantity
avg_cost
created_at
Imports / loads
market data
1 to many
transactions
transaction_id (PK)
portfolio_id (FK)
security_id (FK)
transaction_type
quantity
price
transaction_date
notes
created_at
many to 1
Portfolio uses benchmark
benchmark_history
benchmark_id (PK)
benchmark_name
benchmark_date
close_value
created_at
Imports / loads
benchmark data
data_import_runs
import_run_id (PK)
source
file_name
import_date
status
records_processed
records_failed
error_message
created_at

THANK YOU